Traditional Chinese Medicine Multi-dimensional Knowledge Graph (TCM-MKG) V 1.0

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Abbreviations

Chinese Herbal Pieces (CHP), Chinese Patent Medicines (CPM), Concept Unique Identifiers (CUI), Disease Ontology (DO), Disease Ontology Identifiers (DOID), Graph Attention Network (GAT), Graph Autoencoder (GAE), Graph Convolutional Network (GCN), Graph Neural Network (GNN), Natural Products (NP) , Unified Medical Language System (UMLS), International Classification of Diseases 11 (ICD-11), Medical Subject Headings (MeSH), National Center for Biotechnology Information (NCBI), National Medical Products Administration (NMPA), Protein-Protein Interactions (PPI), Simplified Molecular Input Line Entry System (SMILES), Traditional Chinese Medicine (TCM), Traditional Chinese Medicine Multi-dimensional Knowledge Graph (TCM-MKG), World Health Organization (WHO).

Basic Information of TCM-MKG Data Tables

File Name	Entries
D1_TCM_terminology	1,810
D2_Chinese_patent_medicine	8,977
D3_CPM_TCMT	11,185
D4_CPM_CHP	74,084
D5_CPM_ICD11	69,431
D6_Chinese_herbal_pieces	6,207
D7_CHP_Medicinal_properties	23,517
D8_CHP_NP	6,848
D9_CHP_InChIKey	226,589
D10_Natural_products	4,810
D11_NP_InChIKey	426,305
D12_InChIKey	123,647
D13_InChIKey_EntrezID	2,175,154
D14_InChIKey_SourceID	134,190
D15_InChIKey_distance	123,647
D16_Protein_protein_interactions	6,879,008
D17_Target_Symbol_Mapping	20,053
D18_ICD11	18,444
D19_ICD11_CUI	65,758
D20_ICD11_MeSH	6,853
D21_ICD11_DOID	4,803
D22_CUI_targets	575,153
D23_MeSH_targets	24,097,586
D24_DOID_targets	6,522,131
SD1_predicted_InChIKey_EntrezID	7,551,198

Project Background

Traditional Chinese Medicine (TCM) has a history spanning thousands of years, encompassing systematic theories and practical methods for disease prevention, treatment, and rehabilitation. In contrast to Western medicine's reductionist approach, TCM emphasizes an integrated view of human health, highlighting the dynamic balance and interconnections among various elements such as Qi, blood, Yin-Yang, the organs, and meridians. This holistic perspective has demonstrated unique advantages in addressing complex diseases (e.g., infectious and chronic diseases), especially amid the contemporary global health crisis, which has led to increasing attention to TCM's therapeutic potential and efficacy.

However, TCM research faces significant challenges. First, traditional TCM knowledge is vast and fragmented, lacking a standardized data system. Second, communication between TCM and modern biomedical research remains limited. To better integrate TCM theory with modern scientific methodologies, and to promote its modernization and global application, there is an urgent need to construct a standardized platform capable of integrating multi-dimensional data and supporting interdisciplinary analysis.

To address these challenges, we have developed the **Traditional Chinese Medicine Multi-dimensional Knowledge Graph (TCM-MKG)**. This platform systematically integrates core TCM data and facilitates deep connections with modern biomedical research, providing a powerful tool for interdisciplinary studies and analysis. The aim of this platform is to bridge the data gap in TCM research, advancing the field from traditional experience to modern science, and expanding its application from local research to global use.

Data Open Source and Sharing

The open-source and sharing of TCM-MKG marks a new era of open innovation in TCM research. Through open data, researchers worldwide can seamlessly access and utilize a variety of resources available on the platform, accelerating interdisciplinary collaboration and progress in TCM studies. This open platform not only provides rich research materials for the academic community but also promotes the standardization of TCM, enabling it to integrate seamlessly with modern disciplines such as medicine, pharmacology, and molecular biology.

Moreover, the open-source nature of TCM-MKG lays the foundation for the global application of TCM. Researchers around the world can leverage the platform to conduct cutting-edge studies in areas such as the discovery of new Chinese medicines, disease target research, and personalized treatment design. TCM-MKG's shared platform fosters a deeper understanding and application of

TCM within the global scientific community, advancing the scientific and data-driven integration of traditional medical knowledge with modern technology, and further elevating TCM's role in global health.

TCM-MKG Overview

Core Objectives

Multi-dimensional Data Integration and Systematic Presentation

One of the core goals of TCM-MKG is to build a comprehensive and systematic knowledge graph for TCM by integrating multi-dimensional data. The graph consolidates seven core modules: TCM terminology, Chinese patent medicines (CPM), Chinese Herbal Pieces (CHP), Natural Products (NP), chemical components, targets, and diseases. This integration provides strong data support for interdisciplinary, system-based research. Researchers can use these modules to analyze TCM's mechanisms and potential from diverse disciplinary perspectives.

Promoting Interdisciplinary Innovative Research

TCM-MKG is not only a data platform but also a tool for promoting interdisciplinary collaboration. The exchange between TCM and fields such as modern biomedicine, chemistry, and pharmacology is often hindered by disciplinary barriers. TCM-MKG overcomes these barriers through standardized and systematized knowledge graphs, providing a common research platform for scientists across various disciplines. For example, the correlations between TCM formulas, medicinal herb chemical components, targets, and integrated Chinese-Western medicine treatments can all be deeply explored through this platform.

Standardization and Global Research Foundation

The construction of TCM-MKG adheres to international data processing standards and is integrated with global medical classification systems such as International Classification of Diseases 11 (ICD-11), Unified Medical Language System (UMLS), Medical Subject Headings (MeSH), and Disease Ontology Identifiers (DOID). This ensures that the data is internationally comparable and supports cross-national interoperability. This global framework not only provides a global perspective for TCM research but also facilitates the integration of TCM with the latest developments in modern medicine and global public health. Through this standardized framework, TCM-MKG aids the scientific application of TCM worldwide, contributing to solutions for global health challenges.

Key Features

Modular Design and Systematic Correlation

TCM-MKG consists of seven core modules, each independent in terms of content and function, yet interconnected through refined linkage mechanisms. The TCM terminology module provides the semantic foundation for the entire graph, ensuring seamless data integration across other modules. The CHP, medicinal herbs, and NP modules provide detailed information on therapeutic formulas, drug sources, and chemical components, while the targets module helps researchers analyze the mechanisms of drug molecules and their relationship with diseases. The diseases module, through its connections to other modules (e.g., chemical components and targets), enables the application of TCM within the framework of modern medicine.

Innovative Knowledge Graph Architecture

TCM-MKG's innovation lies in its use of knowledge graph technology, combining graph data structures and semantic reasoning methods to facilitate in-depth analysis of complex TCM data through graph-based intelligent techniques. By applying graph association rules, researchers can discover direct relationships between TCM elements and uncover potential, complex, multi-dimensional connections. For example, graph intelligence algorithms allow researchers to explore latent connections between drug components and diseases, further revealing TCM's potential efficacy and mechanisms.

International Standard Integration and Interoperability

The standardized design of TCM-MKG ensures high compatibility on a global scale. Through integration with international medical classification systems such as ICD-11 and MeSH, TCM-MKG ensures the international applicability of its data, providing a unified analysis framework for researchers worldwide. Whether scholars in TCM or modern biomedicine, all can use this data within a common framework, fostering knowledge sharing and innovation across disciplines.

Structure and Relationships

As illustrated in **Figure 1**, TCM-MKG integrates TCM theory with modern biomedical knowledge to construct a multi-dimensional, complex network encompassing several core modules: TCM terminology, CPM, CHP, NP, chemical components, targets, and diseases. These modules are intricately interconnected, forming a systematic data ecosystem that supports in-depth research and interdisciplinary applications in TCM.

The **TCM Terminology** module provides the semantic foundation for the entire knowledge graph. By standardizing core concepts widely used in the field of TCM, TCM-MKG ensures terminological consistency across modules and provides a basis for data interoperability. This module encompasses

critical TCM concepts such as etiology, pathogenesis, and disease classifications, and explicitly defines the semantic relationships between modules for efficient data linkage.

Building on this foundation, the **CPM** module systematically records the composition, symptoms, pathogenesis, and treatment methods of CHP formulations. It also directly interfaces with modern disease classification systems, such as ICD-11, promoting research and practice in integrative medicine. This module provides rich information on TCM formulas, enabling researchers to analyze the therapeutic potential of CHP within the context of modern diseases.

The **CHP** module extends the scope to include the medicinal properties, efficacy, and chemical components of herbal decoctions. In collaboration with the **NP** module, TCM-MKG not only traces the origins of Chinese herbs but also provides detailed compositional data. The NP module records the biological sources and classification of Chinese herbs, offering insights into their chemical composition and supporting the analysis of their biological characteristics from a phylogenetic perspective.

The **Chemical Components** module standardizes the identification of drug molecules using identifiers such as InChIKey and SMILES, and links chemical components to biological target data via identifiers like EntrezID. This enhances the precision of molecular-level analyses of drug components. It contains extensive chemical component data, categorized into various classes such as terpenoids and alkaloids, providing robust support for drug screening and pharmacological research.

The **Targets** module integrates data on biological targets related to TCM ingredients, with a particular focus on protein-protein interactions (PPI) and the mapping of target genes. This module offers researchers tools to better understand the mechanisms of action of TCM components, further supporting applications in precision medicine and targeted therapy.

The **Diseases** module connects TCM ingredients and targets with modern disease classification systems such as ICD-11, MeSH, and DOID. This multi-dimensional mapping system forms the foundation for interdisciplinary research and application, facilitating the integration of TCM theory into the context of modern medicine.

Overall, TCM-MKG, through the integration of information from the **CPM** module, bridges the gap between traditional TCM concepts such as etiology, pathogenesis, and treatment methods, and modern medical disease classification systems. This connection forms a bridge between traditional and Western medicine. Additionally, the **CHP**, **NP**, and **Chemical Components** modules, through standardized chemical identifiers (e.g., InChIKey), interface with modern chemical databases, providing a solid data foundation for exploring the molecular mechanisms of Chinese medicine and

drug development. The integration of **Targets** and **Chemical Components** further enhances the analytical capabilities of the knowledge graph, allowing researchers to explore the potential biological targets of herbal ingredients and their applications in modern medicine.

By integrating multi-layered information from CHP, diseases, and targets, TCM-MKG enables researchers to employ graph-based intelligent techniques for complex data reasoning, exploring the mechanisms of TCM efficacy and assessing its potential applications in modern disease treatment. In conclusion, TCM-MKG, through the systematic integration of comprehensive TCM data, provides a highly integrated data platform that offers powerful tools for in-depth research on herbal ingredients, biological targets, and their connections to modern diseases.

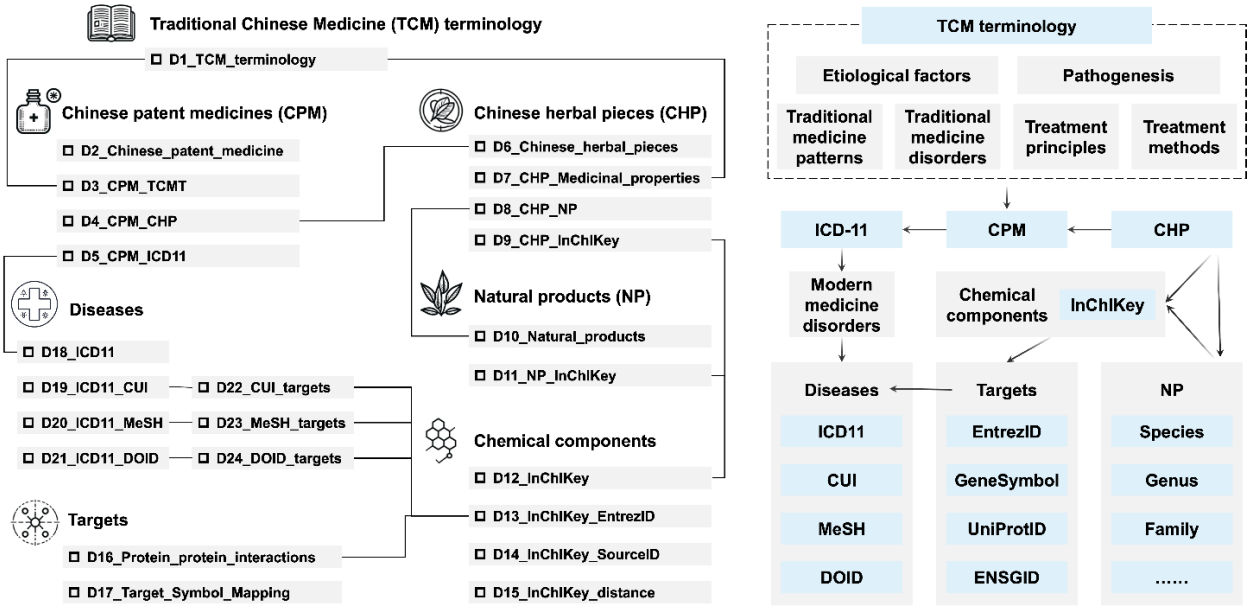


Figure 1: Overview of the Data Structure and Entity Relationships in TCM-MKG

1. TCM Terminology Standard Table (D1_TCM_terminology)

The **TCM Terminology Standard Table** aims to enhance the standardization of TCM terminology and facilitate international communication. By providing detailed term collection and conversion rules, this table enables users to conveniently use and understand TCM terminology, promoting research and practical applications in the field.

1.1 Data Sources

The primary data source is the **WHO International Standard for TCM Terminology** published in 2022 by the World Health Organization (WHO). The publication is available at the following link: https://www.who.int/publications/i/item/9789240042322 (ISBN 9789240042322)

This standard aims to standardize the expression of TCM terminology, particularly for the conversion of synonyms to standard terms, and includes English translations. Additionally, some terms have been constructed internally by the laboratory.

1.2 Function

Standardization of Terms: This module provides a set of standardized TCM terms, reducing discrepancies across various literatures and studies, thus improving communication efficiency.

Synonym Conversion: It provides standardized rules for converting synonyms of TCM terms, ensuring that different expressions are unified into standard terms.

English Translation Standardization: Standard English translations of TCM terms are provided to facilitate international communication and research.

1.3 Basic Structure

The table includes the following fields:

TCMT_ID: A unique identifier for each term.

Chinese_group: The Chinese category to which the term belongs.

English_group: The English category to which the term belongs.

Chinese_term: The standard Chinese term.

Pinyin_term: The Pinyin representation of the Chinese term.

Chinese_synonyms: Synonyms of the standard Chinese term.

English_term: The standard English term.

Synonyms: Synonyms for the English term.

English_definition_description: The English definition of the term.

1.4 Data Scope

This table covers 1810 standard TCM terms, as shown in **Table 1**:

Table 1: D1_TCM_terminology

TCMT_ID	Chinese_group	English_group	Chinese_term	Pinyin_term	Chinese_synonyms	English_term	Synonyms	English_definition_description
TCMT00001	治则	Treatment principles	急则治标		jí zé zhì biāo	Treat the tip first in acute conditions		etc.
TCMT00002	治则	Treatment principles	缓则治本		huǎn zé zhì běn	Treat the root in remissive stages		etc.
TCMT00003	治则	Treatment principles	标本兼治	标本同治	biāo běn jiān zhì	Treat both the tip and root		etc.
TCMT00004	治则	Treatment principles	因时制宜		yīn shí zhì yí	Treat according to time		etc.

2.CPM Basic Information Table (D2_Chinese_patent_medicine)

The CHP Basic Information Table aims to provide essential information and unique identification codes for CHP. Through detailed data collection and standardization, this table facilitates the management and retrieval of CHP information.

2.1 Data Sources

The data is primarily sourced from the 2020 Edition of the Pharmacopoeia of the People's Republic of China and the Chinese Medicine Information Platform (www.dayi.org.cn). This table includes information on CHP approved for sale in China by the National Medical Products Administration (NMPA).

2.2 Function

Basic Information Provision: It provides basic information for each CHP, including name, category, dosage form, etc., which is essential for the database.

Unique Identification Code: Each CHP is assigned a unique code for efficient database management and information retrieval.

Information Standardization: It standardizes the data on CHP, reducing ambiguities in data processing and enhancing the efficiency of data usage.

2.3 Basic Structure

The table includes the following fields:

CPM_ID: A unique identifier for each CHP.

Chinese_patent_medicine: The standard Chinese name of the medicine.

Pinyin_term: The Pinyin representation of the medicine's name.

Routes_of_administration: The routes through which the medicine is administered (e.g., oral, topical).

2.4 Data Scope

This table includes information on 8977 CHP, as shown in Table 2:

Table 2: D2_Chinese_patent_medicine

CPM_ID	Chinese_patent_medicine	Pinyin_term	Routes_of_administration
CPM00001	阿归养血胶囊	Egui Yangxue Jiaonang	Oral
CPM00002	阿归养血颗粒	Egui Yangxue Keli	Oral
CPM00003	阿归养血口服液	Egui Yangxue Koufuye	Oral

3. CPM-TCM Terminology Association Table (D3_CPM_TCMT)

The CHP - TCM Terminology Association Table aims to provide detailed therapeutic methods used in the clinical application of CPM. Through comprehensive data collection and standardization, users can easily query and manage the theoretical information of TCM associated with each CHP.

3.1 Data Sources

The data is sourced from the 2020 Edition of the Pharmacopoeia of the People's Republic of China and the Chinese Medicine Information Query Platform (www.dayi.org.cn). Using the TCM Terminology Standard Table (D1_TCM_terminology), the therapeutic principles and methods for each CHP are standardized, including synonym conversion and Chinese-to-English translation.

3.2 Function

Data Management: This module provides systematic management of therapeutic methods for each CHP, facilitating easy querying and statistical analysis to support research and practical applications in TCM.

Clinical Application Guidance: It offers theoretical insights into the clinical application of each CHP, helping clinicians select and apply Chinese medicines according to TCM principles.

Information Standardization: Synonyms of TCM terminology related to CHP are processed, and both Chinese and English translations are provided to standardize clinical application information and enhance data efficiency.

3.3 Basic Structure

The table consists of the following fields:

CPM_ID: A unique identifier for each CHP.

TCMT_ID: A unique identifier for the corresponding TCM term.

Chinese_group: The Chinese category to which the TCM term belongs.

English_group: The English category to which the TCM term belongs.

Chinese_term: The standard Chinese term used for the therapeutic method.

Pinyin_term: The Pinyin representation of the therapeutic method.

Chinese_synonyms: Synonyms for the standard Chinese term.

English_term: The standard English term.

Synonyms: Synonyms for the English term.

3.4 Data Scope

The table includes 11,185 therapeutic associations between CHP and TCM terms, as shown in Table 3:

Table 3: D3_CPM_TCMT

CHP_ID	TCMT_ID	Chinese_group	English_group	Chinese_term	Pinyin_term	English_term	Synonyms
CHP00002	TCMT00080	治法	Treatment methods	清热解毒	qīng rè jiě dú	Clear heat and remove toxins	Clear heat to detoxify
CHP00003	TCMT01705	传统医学疾病	Traditional medicine disorders	积聚类病	jī jù lèi bìng	Abdominal masses	
CPM00121	Tonify blood						补血

4. CPM-CHP Association Table (D4_CPM_CHP)

The CHP - CHP Association Table focuses on the detailed association between CHP and CHP. Through thorough data collection and standardization, users can easily trace and manage the relationship between CHP and the individual herbal components that they contain.

4.1 Data Sources

The data is sourced from the 2020 Edition of the Pharmacopoeia of the People's Republic of China and the Chinese Medicine Information Query Platform (www.dayi.org.cn). Some CHP, due to confidentiality, may lack specific dosage information for herbal components.

4.2 Function

Association Information Provision: This module provides detailed associations between CHP and their herbal ingredients, including the active parts, processing methods (Pao Zhi), and dosage ratios, which are key components in the database.

Information Traceability: Through the association data, users can trace detailed information about the herbal components in each CHP, facilitating research and practical application.

Data Standardization: It standardizes the association between CHP and herbal pieces, reducing ambiguity in data processing and improving data efficiency.

4.3 Basic Structure

The table includes the following fields:

CPM_ID: A unique identifier for each CHP.

CHP_ID: A unique identifier for each CHP.

Dosage_ratio: The dosage ratio of the herbal piece in the CHP.

4.4 Data Scope

The table includes 74,084 associations between CHP and CHP, as shown in Table 4:

Table 4: D4_CPM_CHP

CPM_ID	CHP_ID	Dosage_ratio
CPM00002	CHP03791	0.0433
CPM00002	CHP05746	0.0433
CPM00002	CHP01717	0.0433
CPM00002	CHP06090	0.0433
CPM00002	CHP00638	0.6968
CPM00002	CHP00641	0.0433
CPM00002	CHP01118	0.0217
CPM00002	CHP03635	0.0433
CPM00002	CHP00419	0.0217

5. CPM-ICD-11 Association Table (D5_CPM_ICD11)

The CHP - ICD-11 Association Table is designed to provide the indications of CHP and their corresponding ICD-11 codes. Through detailed data collection and standardization, this table allows for easy querying and management of clinical application information related to CHP.

5.1 Data Sources

The data is sourced from the 2020 Edition of the Pharmacopoeia of the People's Republic of China and the Chinese Medicine Information Query Platform (www.dayi.org.cn). Furthermore, the International Classification of Diseases 11 (ICD-11) coding tool (https://icd11.pumch.cn/) is used to map CHP indications to their corresponding ICD-11 codes.

5.2 Function

Data Management: This table provides systematic management and query support for clinical application data of CHP, facilitating research and statistical analysis.

Clinical Application Guidance: By linking indications to ICD-11 codes, this table helps clinicians make more informed decisions when selecting CHP, promoting more accurate diagnosis and treatment.

Information Standardization: By associating indications with the internationally recognized ICD-11 codes, this table standardizes the clinical application data of CHP, improving data efficiency.

5.3 Basic Structure

The table includes the following fields:

CPM_ID: A unique identifier for each CHP.

ICD11_code: The ICD-11 code corresponding to the indication for the medicine.

5.4 Data Scope

The table includes 69,431 associations between CPM and ICD-11 codes, as shown in Table 5:

Table 5: D5_CPM_ICD11

CPM_ID	ICD11_code
CPM00009	1F68.1
CPM00009	1F71
CPM00009	2A22

6. CHP Basic Information Table (D6_Chinese_herbal_pieces)

The CHP Basic Information Table provides essential details about CHP, including their names, sources, and synonyms. By standardizing this information, it enables better data management, retrieval, and usage in research and clinical practices related to TCM.

6.1 Data Sources

The data is sourced from the 2020 Edition of the Pharmacopoeia of the People's Republic of China and the Chinese Medicine Information Query Platform (www.dayi.org.cn), covering various types of herbal materials, including animal, plant, and mineral drugs. It also includes different medicinal parts derived from the same natural plant and some processed products.

6.2 Function

Basic Information Provision: Provides foundational details for each herbal piece, including names, sources, and synonyms, which form the core of the database.

Unique Coding: Assigns a unique identifier to each herbal piece to facilitate database management and information retrieval.

Information Standardization: Standardizes the data related to each herbal piece, reducing ambiguity and improving the efficiency of data processing and usage.

6.3 Basic Structure

The table includes the following fields:

CHP_ID: A unique identifier for each CHP.

Chinese_herbal_pieces: The standard Chinese name for the herbal piece.

Chinese_synonyms: Synonyms in Chinese for the herbal piece.

Pinyin_term: The Pinyin representation of the herbal piece name.

English_term: The standard English name for the herbal piece.

Sources: The source category of the herbal piece (e.g., animal, plant, mineral, microorganism).

6.4 Data Scope

The table includes information for 6,207 different herbal pieces, as shown in Table 6:

Table 6: D6_Chinese_herbal_pieces

CHP_ID	Chinese_herbal_pieces	Chinese_synonyms	Pinyin_term	English_term	Sources
CHP00001	阿尔泰多榔菊		a er tai duo lang ju	Doronicum altaicum	Viridiplantae
CHP00002	阿氏蒿		a shi hao	Ashurbajev Wormwood*	Viridiplantae
CHP00003	阿魏	阿里红|臭阿魏	a wei	Resina Ferulae	Viridiplantae

7. CHP Medicinal Properties Table (D7_CHP_Medicinal_properties)

The CHP Medicinal Properties Table provides detailed medicinal properties for each herbal piece. This data reveals their therapeutic value within the TCM system and offers valuable insights for modern scientific research.

7.1 Data Sources

The data is sourced from the 2020 Edition of the Pharmacopoeia of the People's Republic of China and the Chinese Medicine Information Query Platform (www.dayi.org.cn). The medicinal properties include nature, taste, and meridian tropism, which are systematically encoded through network analysis.

7.2 Function

Medicinal Properties Information Provision: Provides medicinal properties for each herbal piece, such as its nature, taste, and meridian tropism, which are key components in the database.

Information Standardization: Standardizes the medicinal properties' data by providing Chinese-English translations and synonym processing, improving the consistency and efficiency of data use.

Data Management: Systematically manages medicinal property data, supporting research and applications in TCM.

7.3 Basic Structure

The table includes the following fields:

CHP_ID: A unique identifier for each herbal piece.

Medicinal_properties: The medicinal properties of the herbal piece (e.g., nature, taste, meridian tropism).

Class: The classification of the medicinal properties (e.g., nature, taste, meridian tropism).

x_rank: The encoded ranking for the medicinal property on the x-axis (e.g., nature).

y_rank: The encoded ranking for the medicinal property on the y-axis (e.g., taste).

7.4 Data Scope

The table includes 23,517 records of medicinal properties for CHP, as shown in Table 7:

Table 7: D7_CHP_Medicinal_properties

CHP_ID	Medicinal_properties	Class	x_rank	y_rank
CHP00001	Sweet medicinal	Medicinal flavor	13	19
CHP00001	Bitter medicinal	Medicinal flavor	10	13
CHP00001	Lung meridian	Meridian tropism	5	8

8. CHP - NP Association Table (D8_CHP_NP)

The CHP - NP Association Table focuses on the relationship between CHP and their NP sources. This data helps trace the origin of each herbal piece, providing a basis for phylogenetic analysis.

8.1 Data Sources

The data is sourced from the 2020 Edition of the Pharmacopoeia of the People's Republic of China and the Chinese Medicine Information Query Platform (www.dayi.org.cn). The NP sources are identified using National Center for Biotechnology Information (NCBI) Taxonomy (https://www.ncbi.nlm.nih.gov/taxonomy)[1], which converts species' Latin names into unique identifiers.

8.2 Function

Source Clarification: This table clearly defines the source for each herbal piece, including one-to-one, one-to-many, and many-to-many relationships between herbal pieces and their sources.

Phylogenetic Analysis: Using the associations, we can conduct phylogenetic studies to explore the evolutionary relationships and classification status of different herbal pieces.

Information Standardization: The use of NCBI Taxonomy identifiers ensures standardized data that facilitates international communication and data sharing.

8.3 Basic Structure

The table includes the following fields:

CHP_ID: A unique identifier for each herbal piece.

Sources: The source category of the herbal piece (e.g., animal, plant, microorganism, mineral).

species_name: The Latin name of the species associated with the herbal piece.

species_ID: The NCBI species identifier for the corresponding species.

8.4 Data Scope

The table includes 6,848 associations between herbal pieces and their NP sources, as shown in Table 8:

Table 8: D8_CHP_NP

CHP_ID	Sources	species_name	species_ID

CHP00001	Viridiplantae	Doronicum altaicum	118749
CHP00002	Viridiplantae	Artemisia adamsii	401875
CHP00003	Viridiplantae	Ferula sinkiangensis	1271632

9. CHP - Chemical Components Association Table (D9_CHP_InChIKey)

The CHP - Chemical Components Association Table aims to provide detailed chemical component information for each herbal piece. This table includes standardized chemical identifiers and integrates data from multiple authoritative sources to facilitate comprehensive chemical research and analysis.

9.1 Data Sources

The data comes from multiple authoritative databases, with chemical components standardized using the RDKit chemical informatics platform (https://www.rdkit.org/) to generate Simplified Molecular Input Line Entry System (SMILES) strings and calculate unique InChIKey identifiers. The sources include:

TCMID (Traditional Chinese Medicine Integrated Database) V2.0 (http://47.100.169.139/tcmid)[2]

TCMbank V1.0 (https://tcmbank.cn)[3]

SymMap V2.0 (http://www.symmap.org)[4]

HIT (Herbal Ingredients Targets) V2.0 (http://hit2.badd-cao.net)[5]

HERB V2.0 (http://47.92.70.12)[6]

ETCM (Encyclopedia of Traditional Chinese Medicine) V2.0 (http://www.tcmip.cn/ETCM2)[7]

TCMSP (Traditional Chinese Medicine Systems Pharmacology Database and Analysis Platform) V1.0 (https://www.tcmsp-e.com/)[8]

9.2 Function

Chemical Component Information: This table records detailed information about the chemical components found in each herbal piece, including their unique InChIKey identifiers. This helps researchers understand the chemical basis of each herbal piece in depth.

Standardized Data Support: By using the RDKit platform for chemical standardization, the data is highly consistent, making it suitable for international exchange and data sharing.

Multi-Database Integration: The table integrates information from various databases, offering a comprehensive set of data for chemical component research and comparative analysis.

9.3 Basic Structure

The table contains the following fields:

CHP_ID: A unique identifier for each CHP.

InChIKey: The unique identifier for each chemical component, calculated using RDKit.

Source: The database from which the chemical component information originates.

9.4 Data Scope

The table includes 226,589 records of chemical component information for CHP, as shown in Table 9:

Table 9: D9_CHP_InChIKey

CHP_ID	InChIKey	Source
CHP00003	BDJJRCFSZXNBSM-UHFFFAOYSA-N	HERB2.0
CHP00003	BOPIMTNSYWYZOC-ILWWEHDPSA-N	HERB2.0
CHP00003	BOPIMTNSYWYZOC-ILWWEHDPSA-N	TCMID

10. NP Information Table (D10_Natural_products)

The NP Information Table focuses on providing detailed taxonomic information about NP that serve as the source for CHP. This table supports the study of the phylogenetic relationships of these NP and aids in understanding their roles in the TCM system.

10.1 Data Sources

The data is sourced from NCBI Taxonomy (https://www.ncbi.nlm.nih.gov/taxonomy) [1] and provides detailed taxonomic information about NP associated with the herbal pieces listed in the CHP - NP Association Table (D8_CHP_NP). This includes a range of classification data from kingdom to species level.

10.2 Function

Detailed Taxonomic Information: This table includes detailed classification information, such as kingdom, phylum, class, order, family, genus, and species. These details help users gain a deeper understanding of the sources of CHP and support research.

Standardized Data Support: Using the NCBI Taxonomy classification system ensures that the data adheres to international standards, facilitating global data exchange and communication.

Phylogenetic Analysis: The detailed taxonomic information allows for phylogenetic analysis, helping researchers explore evolutionary relationships and the classification status of different NP.

10.3 Basic Structure

The table contains the following fields:

kingdom_Name: The name of the kingdom.

kingdom_ID: The ID of the kingdom.

phylum_Name: The name of the phylum.

phylum_ID: The ID of the phylum.

class_Name: The name of the class.

class_ID: The ID of the class.

order_Name: The name of the order.

order_ID: The ID of the order.

family_Name: The name of the family.

family_ID: The ID of the family.

genus_Name: The name of the genus.

genus_ID: The ID of the genus.

species_name: The name of the species.

species_ID: The ID of the species.

10.4 Data Scope

The table includes 4,810 records of taxonomic classification information for NP, as shown in Table 10:

Table 10: D10_Natural_products

kingdom_Name	kingdom_ID	phylum_Name	phylum_ID	class_Name	class_ID	order_Name	order_ID	family_Name	family_ID	genus_Name	genus_ID	species_name	species_ID
Viridiplantae	33090	Streptophyta	35493	Magnoliopsida	3398	Sapindales	41937	Rutaceae	23513	Citrus	2706	Citrus x limon	2708
Viridiplantae	33090	Streptophyta	35493	Magnoliopsida	3398	Sapindales	41937	Rutaceae	23513	Citrus	2706	Citrus sinensis	2711
Viridiplantae	33090	Chlorophyta	3041	Ulvophyceae	33103	Ulvales	3113	Ulvaceae	3114	Ulva	3118	Ulva pertusa	3120
Viridiplantae	33090	Chlorophyta	3041	Ulvophyceae	33103	Bryopsidales	33104	Codiaceae	3131	Codium	3132	Codium fragile	3133

11. NP - Chemical Components Association Table (D11_NP_InChIKey)

The NP - Chemical Components Association Table focuses on providing detailed chemical component information for NP. By consolidating data from multiple sources and applying standardization techniques, this table helps researchers easily access and manage the chemical components associated with NP.

11.1 Data Sources

In addition to the previously mentioned databases (Lab-curated, TCMID, TCMbank, SymMap, HIT, HERB, ETCM, and TCMSP), this table also integrates data from CMAUP V2.0 (https://www.bidd.group/CMAUP/index.html) [9]. During the data organization process, if a record pertains to a CHP, the mapping is done via the CHP - NP Association Table (D8_CHP_NP). If the record is for a NP (i.e., a Latin name), it is directly recorded.

11.2 Function

Supplementary Chemical Component Information: This table augments the data in the CHP - Chemical Components Association Table (D9_CHP_InChIKey), making chemical component information more comprehensive and detailed.

Standardized Data Support: By using the RDKit platform to standardize chemical components and calculate InChIKey as a unique identifier, the data becomes highly consistent and internationally accessible.

Multi-Database Integration: The table combines information from various databases, providing a rich resource of chemical component data that aids in detailed research and comparison analysis.

11.3 Basic Structure

The table consists of the following fields:

SpeciesID: The identifier for the NP species.

InChIKey: The unique identifier for the chemical component.

Source: The source database for the chemical component information.

11.4 Data Scope

This table includes 426,305 records of chemical component information for NP, as shown in Table 11:

Table 11: D11_NP_InChIKey

SpeciesID	InChIKey	Source
2708	AAWZDTNXLSGCEK-LNVDRNJUSA-N	TCMID
2708	AAWZDTNXLSGCEK-UHFFFAOYSA-N	TCMbank
2708	AAWZDTNXLSGCEK-VNDMDPINSA-N	HERB2.0
2708	AAWZDTNXLSGCEK-ZHQZDSKASA-N	TCMbank

12. Chemical Components Basic Information Table (D12_InChIKey)

The Chemical Components Basic Information Table provides detailed basic information about chemical components. It collects and standardizes data from various sources, making it easy for researchers to query and manage this essential information. This table also includes critical data that will aid in machine learning and predictive modeling.

12.1 Data Sources

The table includes chemical components from D9_CHP_InChIKey and D11_NP_InChIKey, as well as additional high-fidelity target data from DrugBank, TTD, and STITCH. The data is standardized using the RDKit platform, with InChIKey as the unique identifier. The table also contains SMILES, InChI, molecular formulas, and key descriptors for each chemical component.

12.2 Function

Basic Information Provision: This table records detailed information for each chemical component, including SMILES, InChI, Molecular_formula, etc., to support chemical structure analysis.

Standardized Data Support: By standardizing the data using RDKit, it ensures high consistency and facilitates international data exchange.

Structure and Property Analysis: This table provides Murcko scaffolds (both Chiral and Achiral) and key molecular descriptors like QED, MolWt, TPSA, which support both chemical and biological research.

12.3 Basic Structure

The table contains the following fields:

InChIKey: The unique identifier for each chemical component (calculated using RDKit).

SMILES: Simplified molecular input line-entry system representation of the chemical.

InChI: International chemical identifier.

Molecular_formula: The chemical formula of the compound.

Chiral_scaffold: The standard Murcko scaffold including chiral centers.

Achiral_scaffold: The Murcko scaffold without chiral information.

QED: Quantitative estimate of drug-likeness.

MolWt: Molecular weight.

ExactMolWt: Exact molecular weight.

TPSA: Topological polar surface area.

MolLogP: Logarithm of the octanol-water partition coefficient.

NumHAcceptors: The number of hydrogen bond acceptors.

NumHDonors: The number of hydrogen bond donors.

NumRotatableBonds: The number of rotatable bonds in the molecule.

MolMR: Molecular refractive index.

FractionCSP3: The fraction of sp3 hybridized carbons in the molecular skeleton.

12.4 Data Scope

This table contains information on 123,647 chemical components, providing comprehensive data for molecular and structural analysis.

13. Chemical Components - Targets Association Table (D13_InChIKey_EntrezID)

The Chemical Components - Targets Association Table focuses on providing detailed target information for each chemical component. This table is crucial for understanding the biological activity of each component and supports the development of predictive models in drug discovery and biomedical research.

13.1 Data Sources

The data for this table is sourced from four authoritative databases, all of which focus on human targets:

DrugBank V5.1.12 (https://go.drugbank.com) [10]

TTD (Therapeutic Target Database) January 10th, 2024 (https://db.idrblab.net/ttd/)[11]

STITCH V5.0 (http://stitch.embl.de_)[12]

BindingDB April 28th, 2024 (https://www.bindingdb.org/) [13]

After standardizing chemical components with RDKit and converting them to InChIKey, and converting targets to EntrezID, this table stores all the relevant target information associated with the chemical components from the Chemical Components Basic Information Table (D12_InChIKey).

13.2 Function

Target Information Provision: This table records the target information for each chemical component, linking the InChIKey of the component to the EntrezID of the corresponding biological target.

Data Standardization: Using RDKit for standardizing chemical components and ensuring consistency of the data across different sources.

Machine Learning Support: The table consolidates high-quality component-target data, providing a solid foundation for machine learning models aimed at predicting drug-target interactions.

13.3 Basic Structure

The table consists of the following fields:

InChIKey: The unique identifier for the chemical component.

EntrezID: The unique identifier for the biological target.

13.4 Data Scope

The table includes 2,175,154 records of chemical component-target associations, as shown in Table 12:

Table 12: D13_InChIKey_EntrezID

InChIKey	EntrezID
LYGQOCVESZDBMV-UHFFFAOYSA-N	51166
OETBHDOEBPITJA-UHFFFAOYSA-N	51166
VIBFDLHDQBMSOA-UHFFFAOYSA-N	51166
SBYIJFJAUBGBOA-UHFFFAOYSA-N	51166

14. Mapping Numbers of Chemical Components in Other Databases (D14_InChIKey_SourceID)

The table of mapping numbers for chemical components in other databases aims to provide detailed source information about the chemical components. This table not only enhances the accuracy and reliability of the data but also provides a solid foundation for data traceability and information integration.

14.1 Data Source

This section provides the corresponding database identifiers of the chemical components involved in the component-target information from D13_InChIKey_ENTREZID. It ensures that each chemical component can be accurately traced within its source database, facilitating subsequent data updates.

14.2 Purpose

Data Traceability: The table offers the corresponding identifiers for chemical components across various databases, allowing researchers to trace back to the original data sources, thereby ensuring data accuracy and reliability.

Information Integration: By integrating information from multiple databases, the table provides comprehensive data on chemical components, facilitating detailed research and analysis.

Standardized Data Support: The table provides standardized mappings between InChIKey and SourceID, facilitating data management and sharing.

14.3 Basic Composition

InChIKey: The unique identifier for a chemical component.

SourceID: The identifier of the chemical component in the corresponding database.

Source: The source database of the component-target information.

14.4 Data Scope

The table covers specific source information related to chemical component-target associations, with a total of 134,190 entries, as shown in Table 13:

Table 13: D14_InChIKey_SourceID

InChIKey	SourceID	Source
LYGQOCVESZDBMV-UHFFFAOYSA-N	T82668	TTD
OETBHDOEBPITJA-UHFFFAOYSA-N	T82668	TTD
VIBFDLHDQBMSOA-UHFFFAOYSA-N	T82668	TTD
SBYIJFJAUBGBOA-UHFFFAOYSA-N	T82668	TTD

15. Chemical Component Classification and Distance Information Table (D15_InChIKey_distance)

The table of chemical component classification and distance information is dedicated to providing detailed classification and distance data of chemical components.

15.1 Data Source

This table includes the classification results of chemical components calculated using NPClassifier V1.5 (https://npclassifier.ucsd.edu/) [14] from the D12_InChIKey dataset. The table comprises three classification levels: Pathway, Superclass, and Class, with up to three categories per level. Additionally, it contains the IsGlycoside field. SMILES strings were extracted from chemical molecules and converted into molecular graphs. Using the RDKit library, SMILES strings were parsed where each atom in the molecule represents a node and the bonds represent edges. The

features of the nodes are the atomic numbers of the atoms. These molecular graphs were then used for subsequent feature extraction and dimensionality reduction analysis.

We utilized five different methods to extract features from the molecular graphs:

Graph Convolutional Network (GCN): A two-layer graph convolutional network was employed, where each layer applied a graph convolution operation followed by a ReLU activation function. The first layer projected the input features into a 64-dimensional space, while the second layer further transformed these features into a 128-dimensional space for the final embedding. The graph convolution operations for each layer were implemented via the DGL library, and the node aggregation used mean pooling for obtaining the graph-wide embeddings during model training.

Graph Attention Network (GAT): A multi-head GAT with two attention layers, each with four attention heads, was used. Each layer performed graph attention operations where nodes focus on their neighbors based on learned attention coefficients. The outputs from multiple attention heads were concatenated and passed through a ReLU activation function to obtain the node embeddings, with a final embedding dimension of 128.

Graph Autoencoder (GAE): The GAE model was used to learn the latent representations of graphs. The encoder part used graph convolution to embed node features into a 64-dimensional space, while the decoder reconstructed the adjacency matrix of the graph from these embeddings. The 128-dimensional embeddings generated by the encoder served as the feature representation of the graph.

Molecular Descriptors: Molecular descriptors were generated using RDKit, totaling 197 descriptors. The processing steps included handling infinite values and missing data, deleting zero-value and constant columns, and performing Z-score normalization. These normalized data were then projected into two-dimensional space using UMAP for visualization and further analysis.

Molecular Fingerprints: Molecular fingerprints were generated using RDKit by converting molecules into 2048-bit Morgan fingerprints (radius 2). These fingerprints represent the structural features of the molecules and were subsequently dimensionalized using UMAP for low-dimensional embeddings, which facilitated subsequent visualization and clustering analysis.

15.2 Purpose

Detailed Classification Provision: The table records the classification information of each chemical component in detail, helping researchers better understand the properties and functions of compounds.

Data Standardization Support: By using NPClassifier V1.5 to standardize the classification of chemical components, the data achieves high consistency and standardization, facilitating international communication and data sharing.

Multilevel Analysis: The table provides information across three classification levels: Pathway, Superclass, and Class, supporting in-depth analysis of chemical components from multiple perspectives.

Glycoside Information Recognition: The IsGlycoside field helps researchers easily identify and filter glycosidic compounds, supporting related studies.

15.3 Basic Composition

InChIKey: The unique identifier for a chemical component.

Descriptors1, Descriptors2: Features from molecular descriptors.

Fingerprints1, Fingerprints2: Features from molecular fingerprints.

GAT1, GAT2: Features from the Graph Attention Network.

GAE1, GAE2: Features from the Graph Autoencoder.

GCN1, GCN2: Features from the Graph Convolutional Network.

Pathway1, Pathway2, Pathway3: Classification pathways of the chemical component, with multiple categories possible.

Superclass1, Superclass2, Superclass3: Superclass classifications of the chemical component, with multiple categories possible.

Class1, Class2, Class3: Specific classifications of the chemical component, with multiple categories possible.

IsGlycoside: Indicates whether the compound is a glycoside.

15.4 Data Scope

The table covers detailed classification information for 123,647 chemical components. The major categories include:

Terpenoids: A class of NP composed of isoprene units, with various biological activities.

Alkaloids: Nitrogen-containing organic compounds with notable physiological activities, widely found in plants, animals, and microorganisms.

Carbohydrates: Organic compounds composed of carbon, hydrogen, and oxygen, playing key roles in energy storage and structural support.

Shikimates and Phenylpropanoids: Compounds synthesized via the shikimate pathway, widely found in plants and possessing various biological functions.

Polyketides: NP synthesized via the polyketide pathway, exhibiting diverse structures and biological activities.

Amino Acids and Peptides: Amino acids are the building blocks of proteins, and peptides are molecules formed by linking amino acids through peptide bonds.

Fatty Acids: Organic acids with long hydrocarbon chains, essential components of lipids, involved in energy storage, cell structure, and signaling.

Others: A category encompassing compounds that do not fit into the aforementioned classifications.

16. Target Interaction Table (D16_Protein_Protein_Interactions)

The target interaction table is designed to provide detailed information on target interactions. Through comprehensive data collection and standardized processing, users can easily query and manage this information, supporting research in TCM and network biology analysis.

16.1 Data Source

The table includes interaction data for all targets from D13_InChIKey_ENTREZID, covering both direct and causal interactions. The data are sourced from the following databases:

STRING V12.0 (https://cn.string-db.org/) [15]

SIGNOR V3.0 (https://signor.uniroma2.it/) [16]

MINT (May 22, 2024) (https://mint.bio.uniroma2.it/) [17]

IntAct (February 15, 2024) (https://www.ebi.ac.uk/intact/home) [18]

BioGRID V4.4.233 (https://thebiogrid.org/) [19]

16.2 Purpose

Interaction Information Provision: The table records detailed interaction information between targets, including both direct and causal interactions, aiding researchers in understanding the connections and networks of the targets.

Support for Network Biology Research: By providing rich interaction data, the table supports network biology research, revealing the complex relationships and functional mechanisms between targets.

Drug Target Discovery and Validation: The table offers key target interaction data for drug development, supporting the discovery and validation of targets and improving the efficiency and success rate of drug development.

16.3 Basic Composition

EntrezID1: The Entrez ID of the first target in the interaction.

EntrezID2: The Entrez ID of the second target in the interaction.

16.4 Data Scope

The table covers interaction information for targets, with a total of 6,879,008 entries, as shown in Table 14:

Table 14: D16_Protein_Protein_Interactions

EntrezID1	EntrezID2
1	2
1	12
1	13
1	19

17. Target Symbol Mapping Table (D17_Target_Symbol_Mapping)

The target symbol mapping table aims to provide detailed mapping information for targets across different databases.

17.1 Data Source

This table includes four types of target identifiers: UniProtID, GeneSymbol, EntrezID, and ENSGID, along with the protein sequence Sequence from UniProt, enabling the extension and querying of biological information related to compound targets across multiple databases. The data are sourced from the following databases:

Ensembl GRCh37.p13 (http://grch37.ensembl.org/Homo_sapiens/Info/Index) [20]

UniProt Release 2024_02 (https://www.uniprot.org/) [21]

17.2 Purpose

Data Integration and Extension: The table provides mappings of target identifiers across various databases, supporting the extension and querying of biological information related to targets in multiple databases.

Cross-Database Querying: The table offers a convenient interface for cross-database querying, helping researchers obtain additional biological and functional information.

Support for Target Research and Analysis: It serves as a foundational resource for target research and functional analysis, supporting the annotation and biological study of targets.

17.3 Basic Composition

UniProtID: The target identifier in the UniProt database.

GeneSymbol: The gene symbol representing the target's gene name.

EntrezID: The target identifier in the Entrez database.

ENSGID: The target identifier in the Ensembl database.

Sequence: The protein sequence.

17.4 Data Scope

The table covers detailed mapping information for all targets, with a total of 20,053 entries, as shown in Table 15:

Table 15: D17_Target_Symbol_Mapping

UniProtID	GeneSymbol	EntrezID	ENSGID	Sequence
P04217	A1BG	1	ENSG00000121410	/
P01023	A2M	2	ENSG00000175899	/
P18440	NAT1	9	ENSG00000103546	/
P11245	NAT2	10	ENSG00000156006	/

18. ICD-11 Disease Information Table (D18_ICD11)

The ICD-11 Disease Information Table is designed to provide detailed disease information and categorization based on ICD-11 codes. Through standardized data collection and classification, users can easily query and manage disease-related information.

18.1 Data Source

The data is sourced from the Chinese and English versions of ICD-11 (2024) (https://icd11.pumch.cn/). This table provides specific information and groupings for all ICD-11 codes included in the TCM-ICD11 association table (D5_CPM_ICD11).

18.2 Purpose

Disease Information Provision: The table records detailed information for each ICD-11 code, including both Chinese and English terms, helping users understand specific disease details.

Standardized Data Management: By using the standardized ICD-11 coding system, the table ensures the data adheres to international standards, facilitating global data sharing and exchange.

18.3 Basic Composition

ICD11_code: ICD-11 code.

BlockId: Block identifier.

English_term: English term.

Chinese_term: Chinese term.

ClassKind: Category type.

DepthInKind: Depth of the category.

IsResidual: Whether it is a residual category.

ChapterNo: Chapter number.

BrowserLink: Browser link.

isLeaf: Whether it is a leaf node.

Primary tabulation: Main classification.

Grouping1, Grouping2, Grouping3, Grouping4, Grouping5: Grouping information.

Version: Version information.

18.4 Data Scope

This table covers detailed information and groupings for all ICD-11 codes, totaling 18,444 entries.

19. ICD-11 to CUI Mapping Table (D19_ICD11-CUI)

The ICD-11 to CUI Mapping Table aims to provide detailed mapping information between ICD-11 codes and the NCBI UMLS Concept Unique Identifiers (CUI).

19.1 Data Source

The data is sourced from the NCBI UMLS [22] (May 6, 2024), accessible via the following link: https://uts.nlm.nih.gov/uts/umls. The mapped semantic types include the following categories:

Disease or Syndrome

Neoplastic Process

Finding

Injury or Poisoning

Congenital Abnormality

Sign or Symptom

Pathologic Function

Acquired Abnormality

Mental or Behavioral Dysfunction

Anatomical Abnormality

Experimental Model of Disease

Cell or Molecular Dysfunction

19.2 Purpose

Standardized Data Support: This table provides mappings between ICD-11 codes and UMLS CUI, ensuring data consistency and standardization for integration and sharing.

Support for Multiple Semantic Types: The table retains mappings across different semantic types, supporting multi-layered and multidimensional disease research and analysis.

Cross-Database Querying: The table offers an easy interface for cross-database querying, allowing researchers to retrieve more biological and clinical information.

Disease Research and Analysis: It provides foundational data for disease research and analysis, supporting annotation, classification, and diagnosis of diseases.

19.3 Basic Composition

ICD11_code: ICD-11 disease code.

CUI: UMLS Concept Unique Identifier.

Semantic_type: Semantic type.

Term: Semantic content.

19.4 Data Scope

This table covers the CUI mapping information for all ICD-11 disease codes, totaling 65,758 entries, as shown in Table 16.

Table 16: D19_ICD11-CUI

ICD11_code	CUI	Semantic_type	Term
1A00	C0008354	Disease or Syndrome	Cholera
1A00	C0086768	Neoplastic Process	Pancreatic Cholera
1A00	C0481432	Finding	Cholera carrier
1A00	C0275731	Disease or Syndrome	Fowl cholera

20. ICD-11 to MeSH Mapping Table (D20_ICD11_MeSH)

The ICD-11 to MeSH Mapping Table is designed to provide detailed mapping information between ICD-11 codes and MeSH identifiers.

20.1 Data Source

MeSH is a standardized vocabulary for identifying medical and biological terms. The data is sourced from the NCBI UMLS (May 6, 2024) [23], and can be accessed via the following link: https://uts.nlm.nih.gov/uts/umls.

20.2 Purpose

Cross-Database Querying: By mapping ICD-11 codes to MeSH identifiers, the table provides an easy interface for cross-database querying, supporting researchers in retrieving more biomedical information.

Data Standardization: It ensures a standardized mapping between ICD-11 disease codes and MeSH terms, enhancing data consistency and interoperability.

Knowledge Integration: Integrating the ICD-11 and MeSH vocabularies, the table supports more comprehensive knowledge management and information retrieval.

20.3 Basic Composition

ICD11_code: ICD-11 disease code.

MeSH: MeSH identifier.

20.4 Data Scope

The table covers detailed mapping information between ICD-11 codes and MeSH identifiers, totaling 6,853 entries, as shown in Table 17.

Table 17: D20_ICD11_MeSH

ICD11_code	MeSH
MD81.3	D000006
MG3Z	D015746
5C81.1	D000012
JA00	D000022

21. ICD-11 to DOID Mapping Table (D21_ICD11_DOID)

The ICD-11 to DOID Mapping Table provides detailed mapping information between ICD-11 codes and DOID.

21.1 Data Source

This mapping is created by first mapping ICD-11 codes to CUI and MeSH, then further mapping to Disease Ontology (DO) (Release 2024_04) [24]. Data can be accessed via the Disease Ontology website: https://disease-ontology.org/do.

21.2 Purpose

Cross-Database Querying: By providing mappings between ICD-11 codes and DOID, the table helps researchers retrieve disease-related information across various databases.

Data Standardization: Ensures standardized mapping between ICD-11 codes and DOID, improving data consistency and interoperability.

Knowledge Integration: The integration of ICD-11, UMLS, MeSH, and Disease Ontology offers comprehensive disease information and knowledge management support.

21.3 Basic Composition

ICD11_code: ICD-11 disease code.

DOID: Disease Ontology Identifier.

21.4 Data Scope

The table covers detailed mapping information between ICD-11 codes and DOID, totaling 4,803 entries, as shown in Table 18.

Table 18: D21_ICD11_DOID

ICD11_code	DOID
5C81.1	DOID:1386
JA40.0	DOID:9667
JA00	DOID:9667
ED51.0	DOID:3138

22. CUI Classification Disease-Target Association Table (D22_CUI_targets)

The CUI Classification Disease-Target Association Table provides detailed information on the association between diseases and targets.

22.1 Data Source

The data is sourced from DisGeNET V7.0 (https://www.disgenet.org/)[25] and records the association information between CUI-classified diseases and EntrezID targets.

22.2 Purpose

Disease-Target Association Analysis: It offers detailed associations between diseases and targets, helping researchers understand the molecular mechanisms of diseases and identify potential therapeutic targets.

Cross-Database Querying: By mapping CUI and EntrezID, the table facilitates querying and retrieval of related information across multiple databases.

Supports Disease Research and Drug Development: The table provides foundational data for disease research and drug development, supporting areas such as disease classification, diagnosis, and target discovery.

22.3 Basic Composition

CUI: UMLS Concept Unique Identifier.

EntrezID: Entrez database target identifier.

22.4 Data Scope

The table covers association information between CUI-classified diseases and targets, totaling 575,153 entries, as shown in Table 19.

Table 19: D22_CUI_targets

CUI	EntrezID
C1260880	5327
C1260880	5328
C1260880	7422
C1260880	3576

23 MeSH Classification Disease-Target Association Table (D23_MeSH_targets)

The MeSH Classification Disease-Target Association Table aims to provide detailed information on the associations between diseases and targets.

23.1 Data Source

The data is sourced from CTD April 2024 (https://ctdbase.org/)[26], which records the associations between diseases classified by MeSH and EntrezID.

23.2 Function

Disease-Target Association Analysis: Provides detailed information on the associations between diseases and targets, helping researchers understand the molecular mechanisms of diseases and potential therapeutic targets.

Cross-Database Queries: Supports cross-database queries by mapping MeSH and EntrezID, allowing researchers to access related information from multiple databases.

Supports Disease Research and Drug Development: Provides foundational data for disease research and drug development, supporting disease classification, diagnosis, and target discovery.

23.3 Basic Components

MeSH: MeSH Concept Unique Identifier.

EntrezID: Entrez database identifier for the target.

23.4 Data Scope

The table includes the associations between MeSH-classified diseases and targets, totaling 24,097,586 entries, as shown in Table 20:

Table 20: Example Data of MeSH Classification Disease-Target Association Table (D23_MeSH_targets)

MeSH	EntrezID
D015746	1
D020434	1
D000022	1
D000138	1

24 DOID Classification Disease-Target Association Table (D24_DOID_targets)

The DOID Classification Disease-Target Association Table aims to provide detailed information on the associations between diseases and targets.

24.1 Data Source

The data is sourced from Diseases V2.0 (https://diseases.jensenlab.org/)[27], which records the associations between diseases classified by DOID and EntrezID.

24.2 Function

Disease-Target Association Analysis: Provides detailed information on the associations between diseases and targets, helping researchers understand the molecular mechanisms of diseases and potential therapeutic targets.

Cross-Database Queries: Supports cross-database queries by mapping DOID and EntrezID, allowing researchers to access related information from multiple databases.

Supports Disease Research and Drug Development: Provides foundational data for disease research and drug development, supporting disease classification, diagnosis, and target discovery.

24.3 Basic Components

DOID: Disease Ontology Identifier.

EntrezID: Entrez database identifier for the target.

24.4 Data Scope

The table includes the associations between DOID-classified diseases and targets, totaling 6,522,131 entries, as shown in Table 21:

Table 21: D24_DOID_targets

DOID	EntrezID
DOID:0050959	6315
DOID:1441	6315
DOID:0050951	6315
DOID:0050753	6315

25 TCM Component-Target Discovery Based on Neighbor Diffusion (SD1_predicted_InChIKey_EntrezID)

In order to improve the target coverage of TCM components, a neighbor diffusion-based method combined with Graph Neural Network (GNN) models is used to predict potential targets of TCM components.

25.1 Data Source

The data is sourced from two files: D15_InChIKey_distance and D13_InChIKey_EntrezID. On this basis, two molecular descriptors (Descriptors1 and Descriptors2), two molecular fingerprints (Fingerprints1 and Fingerprints2), and three GNN models (GCN, GAT, and GAE) were used for dimensionality reduction. Using the get.knnx function from the FNN package, we calculated the distance between each TCM component and the nearest samples with targets in the dataset to ensure the accuracy of the expanded target information.

To avoid self-association interference, records with the same InChIKey were removed, resulting in a set of TCM component-target candidate pairs based on the neighbor diffusion strategy.

For target prediction evaluation, the PSICHIC model (https://github.com/huankoh/PSICHIC) [28] was used to calculate the binding affinity of the expanded targets, validating the reliability of target predictions.

25.2 Function

Target Prediction: By using a neighbor diffusion-based strategy combined with GNN models, this research predicts the potential targets of TCM components, providing support for subsequent pharmacological mechanism studies.

Target Binding Affinity Calculation: The PSICHIC model is used to calculate the binding affinities of expanded targets, evaluating the affinity of predicted targets to improve the accuracy of target predictions.

25.3 Basic Components

InChIKey: International Chemical Identifier, used to uniquely identify chemical molecular structures, ensuring accurate differentiation of TCM components in the data.

EntrezID: Identifier in the Entrez Gene database, pointing to specific gene or protein targets, which is fundamental for target prediction and validation.

Predicted_binding_affinity: The predicted binding affinity of the target, indicating the strength of the binding between the TCM component and the target. Lower binding affinity values indicate stronger binding.

25.4 Data Scope

The TCM component-target discovery information based on the neighbor diffusion method contains 7,551,198 entries, as shown in Table 22:

Table 22: SD1_predicted_InChIKey_EntrezID

InChIKey	EntrezID	predicted_binding_affinity
AAACGARPLFRPEK-MQRLOZBDSA-N	13	5.623248
AAACGARPLFRPEK-MQRLOZBDSA-N	23	5.628702
AAACGARPLFRPEK-MQRLOZBDSA-N	136	9.538736
AAACGARPLFRPEK-MQRLOZBDSA-N	146	9.183507

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